Auditable Adaptive Learning for Windows Malware Detection: Reliability-Aware Fusion, Explanation-Guided Sampling, and TreeSHAP Validation

Luu Van Doanh and Truong Ninh Thuan*

November 1, 2026

Abstract

Windows malware detection requires decisions from incomplete evidence while expert labels remain scarce. Static scores provide early coverage, whereas dynamic API prefixes supply complementary observations after execution; the resulting models also require explanations that can be audited before they guide data acquisition. We formulate these requirements as an auditable adaptive- learning framework with three component-level research questions. First, a control-inspired fusion rule maps disagreement between static and dynamic scores to an adaptive weight and ablates proportional, integral, and derivative terms. Second, a pool-based policy combines predictive uncertainty with TreeSHAP directional entropy and attribution variance. Third, controlled deletion and rank-stability tests assess whether TreeSHAP rankings track fitted-model behavior. The components are evaluated on complementary, non-integrated benchmarks: 1,922 paired MalBehavD-V1 samples for fusion and an identity-audited 2,000-sample, 2,381-feature EMBER subset for active learning and fidelity. On the fusion development cohort, Adaptive P reduced false-positive rate from 12.45% to 11.16% relative to fixed weighting (difference -1.28 percentage points; 95% CI $[-2.47, -0.11]$) and improved Brier score, while integral and derivative terms supplied no consistent incremental benefit. Across five outer folds and five seeds, uncertainty plus directional entropy achieved the best ROC/AP learning-curve areas; the full three-signal policy reached a predeclared validation target with 492 labels, 92 (15.8%) fewer than random acquisition. Replacing the top 50 TreeSHAP-ranked features by training means reduced AUC by 0.306729, versus 0.000979 for equal-budget random replacement, with moderate ranking stability. These results support component-wise evidence allocation and model auditing for generic Windows malware; they do not constitute ransomware- specific, external, or end-to-end validation.

Index Terms— active learning, explainable artificial intelligence, LightGBM, malware detection, model auditing, PID control, score fusion, SHAP, TreeSHAP

1 Introduction

Machine-learning malware detectors operate under partial observability. A portable executable (PE) can be inspected statically before execution, providing immediate features from its headers, sections, imports, strings, and byte statistics. This early evidence supports high-throughput screening, but packing, obfuscation, source shift, and novel implementation choices can weaken learned associations [3, 10, 26]. Dynamic analysis reveals API calls and other behavior that a static representation may obscure, but only after a monitored execution has produced enough events. It adds collection cost, depends on the anal-

*The authors are with the VNU University of Engineering and Technology, Vietnam National University, Hanoi, Vietnam.

ysis environment, and can miss behavior that does not manifest during observation [23, 24]. The operational question is therefore broader than choosing one detector: a system must decide how to allocate heterogeneous evidence as it becomes available.

The same allocation problem reappears during model maintenance. New labeled software is expensive because an analyst must inspect only a small fraction of an incoming pool. Uncertainty sampling prioritizes objects near a model boundary [8, 22], yet it compresses the model's evidence into a probability margin. Two samples can have the same output probability while exhibiting different attribution structures: one may have uniformly weak support, whereas another may combine strong positive and negative effects that cancel. Explanation-derived structure can therefore supply a second view of which labels may be informative. Once explanations influence sample selection, however, they should not be accepted from a visualization alone. Their ranked features must be tested against the fitted model, and their sensitivity to training randomness and data partitions must be reported [1, 27].

We organize these requirements as *auditable evidence allocation under partial observability*. The formulation has three connected stages. At the decision stage, an early static score and sequential dynamic evidence must be combined. At the adaptation stage, a restricted labeling budget must be assigned to candidate samples. At the audit stage, the explanation mechanism used in adaptation and model inspection must be subjected to controlled diagnostics. The connection is conceptual and architectural: the same security-learning process must allocate online evidence, offline labels, and trust in explanatory artifacts. It does not imply that all components were evaluated on one dataset.

The first stage draws on proportional–integral–derivative (PID) control. In a physical feedback loop, a PID controller transforms tracking error into an actuation signal: the proportional term responds to present error, the integral term accumulates persistent error, and the derivative term reacts to error change [5]. We transfer this structure cautiously. Detector disagreement replaces physical tracking error, and the controller output changes the relative weight of static and dynamic scores. There is no physical plant or closed-loop stability guarantee. The design is therefore *control-inspired score fusion*. This distinction is important because component ablation, rather than the PID name, must determine whether temporal memory or a derivative term is useful.

The latest paired evidence changes the scientific conclusion. On 1,922 MalBehavD-V1 samples with static metadata and ordered API events, a proportional adaptive rule reduced false-positive rate (FPR) and Brier score relative to validation-selected fixed weighting. A full PID rule obtained ranking performance comparable to logistic stacking and a lower FPR at its selected operating point, but it incurred a small true-positive-rate trade-off. More importantly, PI and PID did not improve consistently over P within the examined 1–20-event horizon. We report this negative result directly and interpret it as evidence-driven simplification: current disagreement is useful in this cohort, whereas accumulated and derivative disagreement have not yet earned additional complexity.

The adaptation and audit stages are evaluated on a separate, identity-audited subset of EMBER [2]. We combine predictive uncertainty with two summaries of signed TreeSHAP contributions: directional entropy, which measures the balance between positive and negative attribution mass, and attribution variance across coordinates. Five acquisition policies are compared under matched initial sets, pools, budgets, and outer folds. We then evaluate TreeSHAP separately over 25 fitted models by replacing sample-specific top-|SHAP| coordinates with training-derived mean or median values and comparing the result with equal-budget random masks. Spearman and Jaccard agreement distinguish seed sensitivity from fold sensitivity.

This component-wise design answers three research questions:

- **RQ1—adaptive fusion:** On paired event-indexed static and dynamic evidence, how do P, PI, and PID fusion compare with average, max-risk, fixed weighting, and learned logistic stacking in discrimination, calibration, and operating-point trade-offs?

- **RQ2—label allocation:** Under matched training-label budgets, do directional entropy and SHAP variance add useful acquisition information beyond random and uncertainty-only selection?
- **RQ3—explanation audit:** Under training-derived mean and median replacement, do top-|TreeSHAP| coordinates affect fitted- model ranking more than equal-budget random coordinates, and how stable are the rankings across model seeds and data folds?

The study makes four contributions.

- We formulate one auditable evidence-allocation problem that connects sequential decision fusion, budgeted model adaptation, and explanation verification while keeping their empirical estimands separate.
- We provide a paired comparison of simple and learned fusion rules on 1,922 samples. Adaptive P improves FPR by 1.28 percentage points and Brier by 0.00306 relative to fixed weighting, with paired-bootstrap intervals that exclude zero. The ablation also shows that I and D do not provide stable incremental value in the measured horizon.
- We evaluate explanation-guided active learning with five outer folds and five seeds. $U + H$ leads ROC/AP area under the learning curve (AULC), $U + V$ reaches the validation target with the fewest labels, and XAI-Full uses 92 fewer training labels than Random. No policy dominates every criterion.
- We test TreeSHAP rankings with controlled mean/median deletion and stability analysis. Top-ranked deletion produces substantially larger AUC decreases than random deletion, while moderate cross-model agreement exposes dependence on the fitted cohort.

The scope is deliberately narrower than a deployment claim. MalBehavD-V1 is used for fusion; EMBER is used for active learning and fidelity. Their metrics are not merged, and no active-learning–fusion interaction is claimed. Both datasets have generic benign/malware labels, so the study does not establish ransomware-specific or zero-day detection. The fusion comparison is development evidence obtained after the original confirmatory analysis on the same cohort; the active-learning and fidelity results are confirmatory under the retained protocol. These evidence levels are visible in the tables and discussion.

The remainder of the paper is organized as follows. Section 2 reviews static/dynamic detection, score fusion, active learning, and explanation validation. Section 3 states the unified problem and claim boundaries. Section 4 describes the fusion, acquisition, and audit methods. Section 5 separates the two benchmark protocols and evidence tiers. Section 6 reports results by research question. Sections 7 and 8 interpret the findings and threats to validity, and Section 9 concludes.

2 Related Work

2.1 Static and Dynamic Windows Malware Detection

Static malware analysis derives evidence without running a suspect object. Learned detectors have used engineered binary attributes [20], complete byte streams [18], and documented PE representations such as EMBER [2]. SOREL-20M expanded the scale of static PE research and supplied richer metadata [11]. Gradient-boosted trees remain effective on heterogeneous tables containing histograms, strings, imports, sections, and header fields [13]. Their principal operational benefit is early scoring, but static associations can be changed through packing, obfuscation, build-tool variation, and distribution shift [10, 26].

Dynamic analysis instead observes execution. API calls, file and registry operations, process creation, and network activity can expose behavior absent from a static vector [23,24]. The evidence is conditional on the sandbox, inputs, run duration, and path reached by the program. Delayed or environment-aware behavior can remain unobserved, while longer collection increases resource cost. Work on ransomware illus-

trates the value of behavior-centric analysis [6, 14], but the present datasets contain generic malware labels. Those studies motivate the analysis modes; they do not transfer ransomware semantics to our labels.

Hybrid systems commonly concatenate features, combine decisions, or fuse scores. A concatenated model assumes that all modalities are available at once, whereas sequential monitoring produces dynamic prefixes over time. Fixed score averaging is transparent but cannot respond to sample-specific disagreement. Learned stacking can capture a calibrated relation between channels, but it requires a representative development cohort and can obscure how the weight changes as evidence arrives. Meaningful evaluation must compare adaptive rules with simple average, max-risk, fixed weighting, and a learned stacker using the same paired samples and operating-point procedure.

2.2 From Physical PID Control to Score Fusion

PID control is a foundational feedback mechanism. Given reference $r(t)$, measured output $y(t)$, and tracking error $e(t) = r(t) - y(t)$, proportional action responds to present error, integral action addresses persistent accumulated error, and derivative action anticipates rapid error change [5]. These roles are meaningful in a physical loop with a plant, time scale, actuation, and stability analysis.

Mapping PID structure to classifier fusion requires narrower language. Two detector outputs are measurements, not a reference and controlled state. Changing their weight does not alter the program being observed or close a physical loop. In our formulation, static–dynamic disagreement is a surrogate error and the controller output is a bounded mixture weight. The P/I/D terms then provide hypotheses about useful memory: P reacts to current disagreement, I represents persistent disagreement across prefixes, and D represents a change in disagreement. No stability guarantee follows from this analogy. Validation and component ablation decide whether any term improves prediction.

This framing addresses a common methodological gap. A controller name can make a fusion rule appear inherently adaptive, even when each sample provides one observation or the derivative rarely changes a decision. We retain P/PI/PID as nested, testable models and report a negative I/D result. The contribution is therefore an auditable translation and empirical simplification of a control-inspired rule, rather than a claim that classical control theory proves malware-classification performance.

2.3 Active Learning and Batch Acquisition

Pool-based active learning alternates model fitting, query selection, oracle annotation, and retraining. Uncertainty sampling is widely used because it is simple and model compatible [8, 22]. In batch settings, however, the highest-scoring samples can be redundant. Core-set acquisition emphasizes representation coverage [21]; ensemble methods use predictive disagreement [7]; BatchBALD uses joint mutual information [15]; and BADGE clusters uncertain gradient embeddings [4].

These methods are not equally portable to boosted trees. BatchBALD and BADGE were designed around Bayesian or differentiable deep models, although their principles motivate tree-compatible diversity and committee baselines. Our measured policies include random sampling, uncertainty, two explanation-signal ablations, and a three-signal combination. They do not include uncertainty plus k-center diversity or a calibrated committee. We consequently test whether the specific TreeSHAP summaries add value to the retained baselines, without claiming general dominance over active learning.

Explanations may enter active learning as annotator-facing context or as an algorithmic ranking signal. Explainable active learning has studied how local explanations affect annotator experience [9]. Our use is algorithmic: signed TreeSHAP summaries are computed before annotation and enter the acquisition score. A balanced or dispersed attribution vector is not equal to expected information gain; it is a falsifiable heuristic evaluated through matched learning curves, target attainment, low-FPR summaries, and computation.

2.4 Attribution and Explanation Validation

LIME estimates a local surrogate [19]; SHAP unifies additive feature attribution through Shapley values [17]; and TreeSHAP specializes the calculation to tree ensembles [16]. Explanations can still be insensitive to model parameters, manipulated, or misread as causal mechanisms [1,25]. Security inputs add domain constraints and adversarial considerations [27].

Deletion tests ask whether changing highly ranked coordinates alters model output. Their conclusion depends on the intervention: independent replacement can move a vector off the data manifold, and a large effect can reflect perturbation magnitude as well as ranking quality [12, 28]. Training-derived mean and median baselines avoid assuming that zero is a semantically neutral PE value, but they do not guarantee an executable-valid binary. Equal-count random masks control the number of replaced coordinates, though they may select features unused by a tree ensemble. Rank stability offers a complementary view by testing sensitivity to model seeds and data folds.

Our novelty claim is correspondingly specific. We do not claim that static– dynamic fusion, PID structure, active learning, or SHAP is individually new. We contribute a scoped evidence-allocation formulation, a paired P/PI/PID fusion ablation that supports simplification, a signed-mass TreeSHAP acquisition family, and quantitative validation of the attribution ranking. The component-wise protocol makes clear which links are empirically measured and which remain an integration hypothesis.

3 Unified Problem Formulation

3.1 Evidence Allocation under Partial Observability

Let i index a PE sample with binary label $y_i \in \{0, 1\}$, where one denotes generic malware. Static representation x_i^{s} is available before execution and produces score p_i^{s}. A monitored run produces an ordered dynamic sequence $z_i = (z_{i1}, \ldots, z_{iT_i})$. At prefix k, dynamic model g produces $p_{ik}^{\mathrm{d}} = g(z_{i1:k})$. The online task is to map the available evidence to a fused score

$$p_{ik}^{\mathrm{f}} = F\big(p_i^{\mathrm{s}}, p_{i1:k}^{\mathrm{d}}; \theta_F\big), \tag{1}$$

and a decision at a validation-selected threshold. Because k counts API events, it is an observation index rather than elapsed seconds.

The development task has a different resource constraint. Given labeled set $\mathcal{L}_r$, unlabeled pool $\mathcal{U}_r$, and batch budget b, model f_r ranks pool samples with acquisition function Q_r. The oracle reveals labels only for selected batch $\mathcal{B}_r$:

$$\mathcal{B}_r = \operatorname*{arg\,top}_{\substack{\mathcal{B}\subset\mathcal{U}_r \\ |\mathcal{B}|=b}} \sum_{x\in\mathcal{B}} Q_r(x), \qquad \mathcal{L}_{r+1} = \mathcal{L}_r \cup \mathcal{B}_r. \tag{2}$$

The retained experiment uses a pointwise top-b rule; it does not optimize joint diversity.

The audit task operates on a fixed fitted model and explanation $\boldsymbol{\phi}_i$. It tests whether coordinates ranked highly by $|\boldsymbol{\phi}_i|$ produce greater model-performance degradation under a specified replacement intervention than equal-count random coordinates. It also measures ranking agreement across fitted models. This task estimates relative model reliance and sensitivity, not causal behavior of the executable.

These three tasks share an allocation principle but not one objective function. Online fusion allocates weight between evidence channels; active learning allocates labeling budget; explanation audit allocates confidence to a ranked attribution. Reporting one metric as proof of all three would conflate distinct estimands. We therefore use one architecture and three RQs, with separate data and evaluation units.

3.2 Fusion Estimands

Fusion is assessed along three axes. ROC–AUC and average precision (AP) measure ranking. The Brier score

$$\mathrm{Brier} = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2 \tag{3}$$

measures squared probabilistic error. FPR and TPR measure decisions at thresholds selected using validation data. An FPR comparison is interpreted together with TPR because two independently chosen

Table 1: Positioning of the three components. "Open comparison" identifies an evidence gate rather than a completed experiment.

Area	Representative approach	Strength	Limitation relevant here	Role in this study
Static detection	EMBER/boosted trees [2, 13]	Early, scalable PE scoring	Static distribution shift and manipulation	AL/fidelity benchmark and static fusion channel
Dynamic detection	Sandbox/API behavior [23]	Observes executed behavior	Environment and observation dependence	Event-indexed fusion channel
Fusion	Average/fixed/learned stacking	Transparent or data-adaptive combination	Fixed rules ignore disagreement; stacking can hide sequential behavior	Measured baselines for P/PI/PID
PID control	P/I/D feedback [5]	Interpretable current, accumulated, and rate terms	Classifier fusion has no physical plant or stability guarantee	Control-inspired nested ablation
Active learning	Uncertainty [8]	Simple and strong	Margin omits evidence structure	Measured baseline
Batch diversity	Core-set/committee [7, 21]	Reduces redundancy or measures disagreement	Absent from retained artifact	Open comparison
Explanation guidance	Explainable AL [9]	Adds model rationale to selection/annotation	Explanation signal need not equal information gain	TreeSHAP summaries enter query score
XAI evaluation	Sanity/deletion tests [1, 12]	Tests model sensitivity	Intervention may be off-manifold	Mean/median deletion plus stability

thresholds need not correspond to the same operating point.

Paired differences use identical test rows. For Adaptive P versus Fixed, the primary operating-point effects are ΔFPR, ΔTPR, and ΔBrier. For PID versus learned stacking, the report additionally uses ΔAUC. Bootstrap intervals quantify paired uncertainty over the retained cohort. The post-confirmatory fusion development used this same cohort; its intervals describe resampling uncertainty, not external confirmation.

3.3 Learning-Curve Estimands

Every active-learning run produces ROC–AUC and AP at budgets $B \in \{200, 300, \ldots, 1200\}$. For metric $m(B)$, normalized area under the learning curve is

$$\mathrm{AULC}(m) = \frac{1}{B_{\max} - B_{\min}} \int_{B_{\min}}^{B_{\max}} m(B)\, dB, \quad (4)$$

computed by trapezoidal integration over measured budgets. This quantity rewards useful models earlier in the acquisition process. Endpoint comparison is less informative because every policy eventually receives the same full pool.

A full-pool reference model defines a fold-specific validation target

$$t_{\mathrm{fold}} = 0.5 + 0.95 \left(\mathrm{AUC}_{\mathrm{val}}^{\mathrm{full}} - 0.5 \right). \quad (5)$$

Labels-to-target is the first training budget that reaches this value. It counts queried training labels; the validation labels, original retained labels, and full-pool reference are additional experimental resources. Low-FPR summaries include normalized partial AUC up to FPR 5% and TPR-AULC at FPR caps of 1% and 5%. These are cohort metrics, not estimates for operational FPR near 10^{-3} or below.

3.4 Explanation Object and Intervention

For a fitted tree model and sample $x_i \in \mathbb{R}^d$, let $\boldsymbol{\phi}_i = (\phi_{i1}, \ldots, \phi_{id})$ denote signed TreeSHAP contributions and ϕ_{i0} the bias. Native TreeSHAP is verified on the raw score scale:

$$f_{\mathrm{raw}}(x_i) = \phi_{i0} + \sum_{j=1}^{d} \phi_{ij}. \quad (6)$$

This identity checks numerical reconstruction; it does not verify feature semantics or causal correctness.

For deletion budget k, sample-specific mask $M_i^{(k)}$ replaces selected coordinates with training-derived baseline b_j:

$$\widetilde{x}_{ij} = \begin{cases} b_j, & j \in M_i^{(k)}, \\ x_{ij}, & \text{otherwise.} \end{cases} \tag{7}$$

The top mask contains the k largest $|\phi_{ij}|$ values; the control contains k uniformly sampled coordinates. b_j is the training-column mean or median. The comparison estimates fitted-model sensitivity under two feature-space interventions. A modified vector need not correspond to a buildable or executable PE file.

3.5 Security and Claim Boundaries

The labels in both benchmarks denote generic benign/malware status. Neither dataset establishes ransomware ground truth. The EMBER folds are random row folds, not temporal or family-disjoint splits. MalBehavD-V1 dynamic sequences are event-indexed, not wall-clock traces, and source differences may correlate with class. The study excludes a white-box adversary who manipulates features, pool membership, or the oracle against the complete pipeline.

The unified architecture is consequently a research framework. The fusion component is evaluated on MalBehavD-V1; active learning and fidelity are evaluated on EMBER. There is no measured interaction between adaptive fusion and active learning, no end-to-end latency, and no common cohort supporting a joint performance metric. These boundaries determine the wording of the abstract, results, and conclusion.

4 Methodology

4.1 Decision–Adaptation–Audit Architecture

Figure 1 presents the unified flow. The online decision path computes a static score and, when behavioral evidence is available, updates a dynamic score at ordered event prefixes. A fusion rule returns a final score. The offline adaptation path trains a static model, ranks the remaining pool, asks an oracle for a restricted label batch, and retrains. The audit path freezes a model and tests the attribution ranking through controlled deletion and cross-model stability.

TreeSHAP has two offline roles. It supplies local summaries to the acquisition function and a ranking to the audit experiment. These uses share a fitted explanation method, but their estimands differ: acquisition asks which object to label, while deletion asks whether a ranked coordinate set influences model behavior under a specific intervention. A positive deletion result does not prove that SHAP causes the active-learning improvement.

4.2 Static Learner and TreeSHAP

LightGBM represents a raw score as an additive tree ensemble

$$F(x) = \sum_{m=1}^{M} \eta h_m(x), \tag{8}$$

where h_m is a decision tree, M is the number of boosting rounds, and η is the learning rate [13]. Its histogram-based learner handles heterogeneous PE attributes without forcing a common feature unit. The binary score follows from the logistic link.

Native TreeSHAP returns one signed contribution per feature and a separate bias [16]. Equation (6) is checked before using an explanation. The bias is excluded from feature ranking, directional entropy, variance, and deletion budgets. Including it as a coordinate would confound the model base rate with an observed PE feature.

4.3 Control-Inspired Reliability-Aware Fusion

4.3.1 Translation from classical PID

Classical discrete PID maps tracking error to an actuation signal. Here, the available surrogate is detector disagreement at event prefix k:

$$e_{ik} = p_i^{\mathrm{s}} - p_{ik}^{\mathrm{d}}. \tag{9}$$

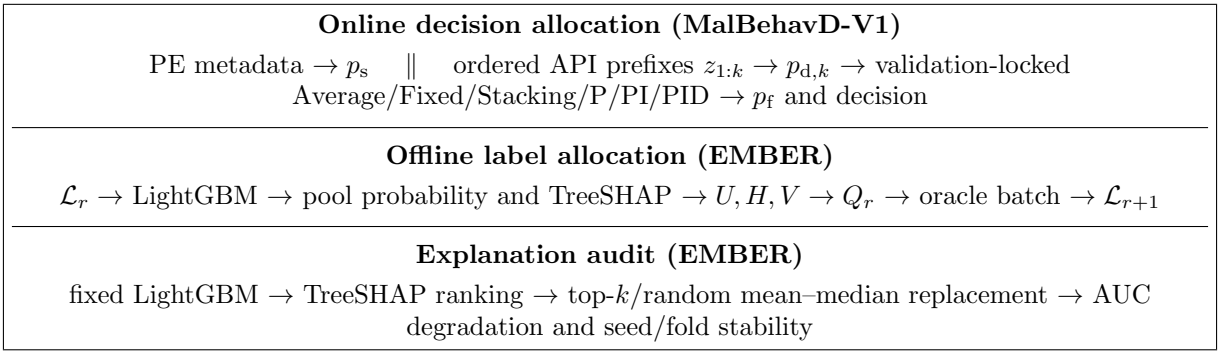


Figure 1: One auditable evidence-allocation formulation with component-wise evaluation. Solid flows show the measured components. The horizontal boundaries mark distinct benchmarks; they are not a single end-to-end experiment.

Positive e_{ik} means that the static channel currently assigns higher malware risk; negative e_{ik} means the dynamic channel assigns higher risk. This choice is interpretable but does not make either detector a reference signal.

For each sample sequence, the state is initialized to zero and updated as

$$I_{ik} = I_{i,k-1} + e_{ik}, \tag{10}$$

$$D_{ik} = e_{ik} - e_{i,k-1}, \tag{11}$$

$$u_{ik} = K_p e_{ik} + K_i I_{ik} + K_d D_{ik}. \tag{12}$$

Controller output is mapped to a bounded static-channel weight:

$$w^{\mathrm{s}}_{ik} = \mathrm{clip}\left(\frac{1}{1+\exp(-u_{ik})}, w_{\min}, w_{\max}\right), \tag{13}$$

and the fused score is

$$p^{\mathrm{f}}_{ik} = w^{\mathrm{s}}_{ik} p^{\mathrm{s}}_i + (1 - w^{\mathrm{s}}_{ik}) p^{\mathrm{d}}_{ik}. \tag{14}$$

The measured implementation bounds the mixture weight; state and threshold are reset or selected without test transfer. An output clip is not equivalent to integral anti-windup, and no Δt appears because the primary sequences are indexed by event count.

The nested family provides the scientific ablation. Adaptive P sets $K_i = K_d = 0$ and responds to current disagreement. PI additionally remembers persistent signed disagreement. Full PID also reacts to the change between adjacent prefixes. If P performs as well as PI/PID, the data do not support the extra memory terms. This is a useful negative conclusion because derivative action can amplify noisy prefix-to-prefix score changes and integral action can accumulate a cohort-specific bias.

4.3.2 Fusion baselines

All measured rules receive the same paired static and dynamic scores. Average is $0.5p^{\mathrm{s}} + 0.5p^{\mathrm{d}}$. Max-risk takes the larger channel score. Fixed weighting selects one constant mixture using validation data. Logistic stacking fits a learned combination on development inputs. P, PI, and PID use the update above, with controller settings and decision threshold locked before test evaluation. The paper compares discrimination, Brier score, and paired FPR/TPR outcomes. A lower FPR at two separately selected thresholds is not called a same-TPR improvement.

4.4 Explanation-Guided Active Learning

4.4.1 Predictive uncertainty

For pool probability p_i, uncertainty is

$$U_i = 1 - 2|p_i - 0.5|. \tag{15}$$

It equals one at the decision boundary and zero at either probability extreme. It is inexpensive and provides the strongest measured baseline beyond random selection.

4.4.2 Directional entropy

For signed attribution vector $\boldsymbol{\phi}_i$, define positive and negative attribution masses

$$A_i^+ = \sum_j \max(\phi_{ij}, 0), \qquad A_i^- = -\sum_j \min(\phi_{ij}, 0). \tag{16}$$

When $A_i^+ + A_i^- > 0$, the positive-mass share is

$$q_i = \frac{A_i^+}{A_i^+ + A_i^-}. \tag{17}$$

Directional entropy is

$$H_i = -q_i \log_2 q_i - (1 - q_i) \log_2 (1 - q_i), \tag{18}$$

with $H_i = 0$ for a zero or one-sided vector. High H_i indicates substantial positive and negative attribution masses in the same prediction. It is not class entropy or calibrated epistemic uncertainty.

Let $S_i = \sum_j \phi_{ij}$ and $L_i = \|\boldsymbol{\phi}_i\|_1$. Since $A_i^+ - A_i^- = S_i$ and $A_i^+ + A_i^- = L_i$,

$$q_i = \frac{1}{2} + \frac{S_i}{2L_i}, \qquad L_i > 0. \tag{19}$$

Thus H_i summarizes signed-mass balance determined by raw-score contribution and attribution magnitude. Future work should compare it with direct margin and norm controls.

4.4.3 Signed-attribution variance

The dispersion signal is

$$V_i = \frac{1}{d} \sum_{j=1}^{d} (\phi_{ij} - \bar{\phi}_i)^2, \qquad \bar{\phi}_i = \frac{1}{d} \sum_j \phi_{ij}. \tag{20}$$

Equivalently,

$$V_i = \frac{\|\boldsymbol{\phi}_i\|_2^2}{d} - \frac{S_i^2}{d^2}. \tag{21}$$

It measures dispersion across coordinates of one explanation, not variance across trees, posterior draws, or independently trained models.

Table 2: Measured acquisition policies.

Policy	(α, β, γ)
Random	not applicable
Uncertainty	$(1, 0, 0)$
$U + H$	$(3/7, 4/7, 0)$
$U + V$	$(0.5, 0, 0.5)$
XAI-Full	$(0.3, 0.4, 0.3)$

4.4.4 Composite acquisition policies

Because U, H, V have different numeric ranges, each signal S is min–max normalized over the current pool:

$$\widetilde{S}_i = \begin{cases} \dfrac{S_i - S_{\min}}{S_{\max} - S_{\min}}, & S_{\max} > S_{\min}, \\ 0, & \text{otherwise.} \end{cases} \tag{22}$$

The general acquisition function is

$$Q_i = \alpha \widetilde{U}_i + \beta \widetilde{H}_i + \gamma \widetilde{V}_i. \tag{23}$$

Table 2 defines the measured policies. Random assigns an independent random score. Ties in deterministic policies use the retained row identifier. Min–max scaling is pool dependent, and the fixed weights were not selected through an independent outer sensitivity study; they are measured configurations rather than established optima.

At each round the algorithm fits a fresh LightGBM model, evaluates validation and held-out test outputs for reporting, computes Q only on the remaining development pool, reveals the top batch labels, and repeats. Test outcomes never affect queries, target construction, or stopping. TreeSHAP is recomputed after each model update, so the acquisition geometry follows the current learner.

4.5 Quantitative TreeSHAP Audit

For $k \in \{10, 25, 50\}$, every test sample receives a top-$|\phi|$ mask and 20 uniform random masks of equal size. The intervention in Equation (7) is repeated with training-column mean and median. The model is not retrained. The primary relative deletion effect is

$$\Delta_k^{\text{rel}} = \Delta\text{AUC}_k^{\text{top}} - \Delta\text{AUC}_k^{\text{random}}. \tag{24}$$

Absolute differences are preferred to ratios because random-mask AUC decreases are near zero.

Global ranking stability is measured by Spearman correlation. Agreement between top-k feature sets is

$$J_k(A, B) = \frac{|T_k(A) \cap T_k(B)|}{|T_k(A) \cup T_k(B)|}, \tag{25}$$

where $T_k(A)$ is the top-k set from model A. Comparisons are separated into different seeds within one fold and models trained on different folds.

The retained artifact also reports explanation consistency (EC), contrastive geometry (CF), and numerical additivity. We keep them descriptive. They have different scales and constructs; column permutations can preserve additivity and much of the geometry while assigning contributions to incorrect feature names. We therefore do not aggregate them into an Overall Fidelity Score.

5 Experimental Protocol

5.1 Component-Wise Evaluation Design

The study uses two complementary benchmarks because no retained cohort contains all required static features, dynamic sequences, active-learning identity, and fidelity artifacts. MalBehavD-V1 answers RQ1. An identity-audited EMBER subset answers RQ2 and RQ3. Table 3 makes their semantics and roles explicit. Results are never pooled across datasets.

Both labels mean generic benign/malware status. AVClass is available for 498 EMBER malware rows but is not used as ransomware ground truth. MalBehavD-V1 may contain source differences between benign and malware collection. The experiment does not infer family-held-out, temporal, or cross-organization performance.

5.2 EMBER Identity, Splits, and Configuration

The EMBER-derived table contains 2,000 unique SHA-256 identities, unique row and source indices, and no exactly duplicated feature vectors. Samples appeared in November–December 2018. The locked dataset SHA-256 is `69f0ae680846edd95458d1e7aaabe36f8866f4f22a6d11488b126ed026b28963`, and the manifest SHA-256 is `a2ca16402653e6cc51fdd7a4857d007f-af86807b044d78654f180409c73565bb`. The subset was recovered from EMBER material but does not reproduce the complete official benchmark protocol.

Five random outer folds create held-out test partitions. Within each outer development partition, retained rows are separated into an acquisition pool and validation resource. Test rows do not enter query selection, target definition, threshold choice, or stopping. All policies in the same fold/seed receive the same initial labeled set and remaining pool. Exact identity auditing reduces row-level leakage, but random folds can still place related families, sources, or time periods on both sides.

Each AL run begins with 200 stratified labels and performs ten rounds of 100 queries, reaching a maximum of 1,200 training labels. Five AL seeds are used per outer fold. The retained LightGBM configuration uses 60 trees, learning rate 0.05, 15 leaves, maximum depth 4, and seed-specific deterministic training. Each round retains pool membership, query components, selected row IDs, model configuration, and validation/test predictions.

For fidelity, each fold fits five LightGBM models with seeds 41–45, 100 trees, learning rate 0.1, feature fraction 0.8, bagging fraction 0.8, 15 leaves, and maximum depth 4. The 25 fitted models are evaluated with mean and median replacement, three deletion budgets, and 20 random masks per budget. Seed comparisons stay within a fold; fold comparisons change the data partition.

5.3 MalBehavD-V1 Pairing and Fusion Protocol

The MalBehavD-V1 source contains 2,570 rows, balanced between 1,285 benign and 1,285 malware entries, with 2,543 unique SHA-256 values. Twenty-six SHA values are duplicated without label conflict. Requiring at least 20 API events yields

Table 3: Dataset provenance, label semantics, observation unit, and evaluation role.

Dataset	Retained samples	Class counts	Inputs / observation unit	Split	Role
MalBehavD-V1	1,922 usable of 1,923 with $\geq$ 20 events	932 benign / 990 malware	199 static metadata attributes and ordered API-event prefixes	5 outer folds; seed 2026	Fusion baselines and P/PI/PID ablation (RQ1)
EMBER-derived identity audit	2,000	1,500 benign / 500 malware	2,381 static PE features; unique SHA-256, row ID, source index	5 random outer folds; 5 AL/model seeds	Active learning (RQ2) and TreeSHAP audit (RQ3)
MalBehavD-V1 horizon-40 subset	1,186	Generic benign/malware	Ordered API-event prefixes through event 40	Development subset	Exploratory robust PID only
Timestamp feasibility pilot	8 usable sequences from 20 selected samples	2 benign / 6 malware usable	4,697 API–timestamp pairs from Cuckoofork reports	No outer-fold evaluation	Parser/data-source feasibility only

1,923 candidates; 1,922 have usable static metadata and dynamic sequences, providing 99.95% metadata coverage. The final paired cohort contains 932 benign and 990 malware samples. Its manifest SHA-256 is `9540b88b0da2463f63b4e983f5ecc277e3845b7badebbb79695e35f2c3b1a669`.

Static input comprises 199 structural metadata attributes. The allow-list excludes verdict, detection count, threat name, tag, and filename fields that could directly encode collection or label information. The dynamic input is an ordered API sequence scored at prefixes. The collector obtains metadata/report records by SHA-256 and does not download or execute PE objects. Event index is the primary observation axis; it cannot support a wall-clock latency claim.

Five outer folds use seed 2026. Model, fusion parameters, fixed weight, stacking, and decision thresholds are selected with development/validation resources and then locked for each held-out evaluation. All fusion methods use the same paired channel scores. Paired bootstrap intervals compare predictions on identical rows. The threshold objective is validation based, so every reported FPR is paired with TPR and is not treated as a common-TPR comparison unless explicitly stated.

The original confirmatory run established channel complementarity and the absence of a useful derivative effect under its protocol. Adaptive P and the new fusion-only comparison were then developed on the same 1,922-sample cohort. Those values are measured on outer-test rows but are classified as development evidence because design choices followed inspection of the confirmatory result. External or temporal validation is required before promoting them to an independent confirmatory claim.

5.4 Metrics and Paired Reporting

Fusion reports ROC–AUC, AP, FPR, TPR, and Brier score. The main table compares Average, Max-risk, Fixed, Logistic stacking, Adaptive P, Adaptive PI, and Full PID. Primary paired effects are Adaptive P minus Fixed and Full PID minus Logistic stacking. An additional P versus PID comparison tests incremental I/D value.

Active learning reports ROC-AULC, AP-AULC, training labels to the fold-specific target, restricted-FPR summaries, and scoped computation. The paired comparison unit is outer fold after averaging its seed trajectories; 25 fold–seed runs are not treated as 25 independent datasets. Query timing includes pool prediction, TreeSHAP where applicable, additivity verification, normalization, and ranking. It excludes human annotation and endpoint feature extraction.

Fidelity reports AUC decrease under top and random deletion, the absolute top-minus-random effect, Spearman rank correlation, and top-k Jaccard agreement. The 20 random masks describe control variation conditional on fitted models; they are not independent deployment datasets. EC, CF, and additivity are secondary diagnostics and are not combined into a composite score.

5.5 Confirmatory versus Exploratory Status

Table 4 locks the evidence hierarchy before interpretation. This prevents a favorable exploratory value from silently replacing a less favorable confirmatory result.

The source-of-truth report identifies experimental commit `8bfa06b65b93`. That commit and the new result directories were not present on the remote branch available to this writing environment. The numerical paper is therefore assembled from the locked readiness report and its artifact map, not described as an independent replay. Before submission, the source commit, artifacts, hash verification, and figure generation must be synchronized in a clean environment.

6 Results

6.1 RQ1: Static–Dynamic Fusion

6.1.1 Overall comparison

Table 5 reports the fusion-only comparison on all 1,922 paired samples. Average produces the highest AUC among the simple nonadaptive rules (0.9840) and the lowest Brier score overall (0.0450). This strong baseline prevents the adaptive family from being judged only against a weak fixed rule. Max-risk performs worst in ranking, FPR, and Brier, showing that always taking the larger risk score amplifies channel errors.

Adaptive P attains the lowest observed FPR, 11.16%, while preserving TPR close to Fixed (95.96% versus 96.06%). PI and PID provide nearly identical AUC/AP and TPR and slightly higher FPR than P. Full PID is therefore not the best member of its own nested family for the selected operating-point objective. The result supports present-disagreement adaptation, not a generic advantage for all three controller terms.

Figure 2 displays the metric trade-off. Average leads AUC and Brier by small margins; Adaptive P leads FPR. Ranking and operating-point objectives select different methods. This is why the paper does not collapse the evaluation into one "best fusion" statement.

6.1.2 Paired effects and controller ablation

Table 6 reports the paired comparisons that govern the claim. Adaptive P reduces FPR by 1.28 percentage points relative to Fixed, with 95% CI $[-2.47, -0.11]$. It also reduces Brier by 0.00306, with CI $[-0.00585, -0.00030]$. The TPR difference is -0.10 percentage point and its interval crosses zero. In counts, P produces 104 false positives versus 116 for Fixed and 950 true positives versus 951. The supported statement is improved FPR and calibration with no detected material TPR loss on this development cohort.

Full PID versus Logistic stacking illustrates a different trade-off. PID lowers FPR by 3.32 percentage points, but the thresholds are selected independently and TPR is 0.62 percentage point lower. Its AUC difference is only +0.08 percentage point with an interval spanning zero. We therefore describe AUC as comparable and avoid a same-TPR or universal-superiority claim.

The nested comparison is decisive for controller complexity. Full PID increases FPR by 0.43 percentage point over Adaptive P while TPR is unchanged. The original confirmatory ablation reached the same qualitative conclusion: turning off K_d while retaining K_p, K_i changed none of the 1,922 final decisions, and D altered only 6 of 7,688 intermediate-prefix decisions before the differences disappeared at the final prefix. PID minus PI AUC was -0.000125 with an interval crossing zero. Thus the retained event horizon does not support a derivative benefit, and the integral term also fails to improve on P.

6.1.3 Answer to RQ1

Current static–dynamic disagreement can usefully adapt the mixture: Adaptive P improves FPR and Brier over Fixed on the measured development cohort. Full PID can achieve AUC comparable to learned stacking and a lower FPR at its own selected operating point, with a small TPR trade-off. The available evidence does not support incremental I/D

Table 4: Evidence hierarchy used throughout the manuscript.

Tier	Artifact family	Permitted interpretation
Confirmatory verification pass	EMBER identity AL/fidelity	Main evidence for RQ2/RQ3; use in abstract and conclusion within dataset scope
Original confirmatory protocol	MalBehavD-V1 1,922	Channel complementarity and component ablation; D does not add stable value
Measured development evidence	Fusion-only 1,922	Main RQ1 comparison; explicitly label same-cohort post-confirmatory development
Exploratory development	Robust PID v2, horizon 40	Hypothesis and algorithm candidate; do not claim external confirmation
Exploratory controlled stress	Stateful gated residual PID	Behavior under specified synthetic delay stress only
Measured feasibility	Timestamp pilot	Data-source/parser feasibility; no performance or D-term conclusion
Excluded	Legacy, simulated, or dashboard values	No use as manuscript evidence

Table 5: Fusion comparison on 1,922 paired MalBehavD-V1 samples. Adaptive P/PI/PID are measured development evidence on the same cohort used during method development. FPR and TPR are reported together at validation-selected operating points.

Method	ROC–AUC	AP	FPR (%)	TPR (%)	Brier
Average 50/50	**0.9840**	0.9880	14.70	**96.77**	**0.0450**
Max-risk	0.9779	0.9794	16.63	**96.77**	0.0604
Fixed weight	0.9821	0.9862	12.45	96.06	0.0487
Logistic stacking	0.9819	0.9867	14.91	96.57	0.0454
Adaptive P	0.9827	0.9867	**11.16**	95.96	0.0456
Adaptive PI	0.9827	0.9867	11.48	95.96	0.0456
Full PID	0.9828	**0.9868**	11.59	95.96	0.0455

complexity, closed-loop stability, sleep- evasion handling, wall-clock latency reduction, or external validation.

6.2 RQ2: Explanation-Guided Label Allocation

6.2.1 Learning-curve quality and target attainment

Table 7 integrates performance across 200–1,200 training labels. $U + H$ achieves the highest ROC-AULC (0.933026) and AP-AULC (0.883336). Relative to Random, its paired fold improvements are +0.004249 ROC-AULC and +0.014985 AP-AULC in the favorable direction in all five folds. Gains over Uncertainty are smaller: +0.001754 and +0.001401, favorable in four of five folds.

$U+V$ reaches the validation target with the fewest mean training labels (488), while XAI-Full reaches it with 492. Relative to Random's 584, XAI-Full reduces the mean training budget by 92 labels, or 15.8%. It also uses 28 fewer labels than Uncertainty, but this advantage does not imply consistent superiority in learning-curve ranking. Figure 3 shows that each explanation-guided policy emphasizes a different criterion.

Table 8 limits the strongest claim. XAI-Full improves over Random in both AULC metrics and all five fold directions, but its AP-AULC difference from Uncertainty is slightly negative and its ROC improvement occurs in only three of five folds. Directional entropy provides the most consistent measured addition; the three-signal policy is not a universal winner.

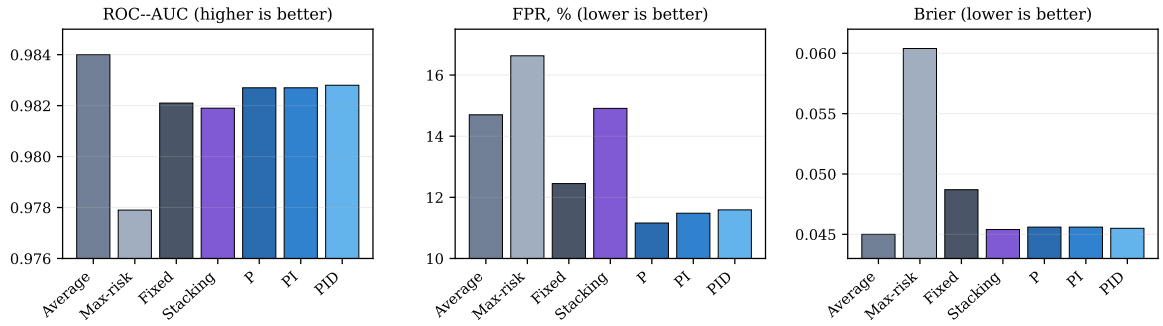


Figure 2: Fusion discrimination, false-positive rate, and Brier score on the paired development cohort. Values are aggregate means supplied by the retained report; error bars are omitted because method-level fold vectors are not present in this writing checkout.

Table 6: Primary paired fusion effects. Percentage-point (pp) effects are method minus comparator. The two methods in a row can use independently validation- selected thresholds.

Comparison	Metric	Difference	95% CI	Interpretation
Adaptive P − Fixed	FPR	−1.28 pp	$[-2.47, -0.11]$	Lower observed false-positive rate
Adaptive P − Fixed	Brier	−0.00306	$[-0.00585, -0.00030]$	Lower squared probability error
Adaptive P − Fixed	TPR	−0.10 pp	includes 0	No detected TPR difference
Full PID − Stacking	FPR	−3.32 pp	$[-4.94, -1.72]$	Lower FPR at separately selected operating point
Full PID − Stacking	AUC	+0.08 pp	$[-0.19, +0.37]$	Comparable ranking; interval includes 0
Full PID − Stacking	TPR	−0.62 pp	$[-1.21, 0.00]$	Small sensitivity trade-off
Full PID − Adaptive P	FPR	+0.43 pp	$[+0.11, +0.86]$	I/D worsen FPR relative to P

6.2.2 Restricted-FPR behavior and computation

Table 9 shows that Uncertainty remains strong in restricted-FPR regions. It leads normalized partial AUC at FPR$\leq$ 5% and TPR-AULC at the 1% cap; XAI-Full leads only at the 5% cap. Explanation-guided policies clearly improve over Random, but their differences from Uncertainty are small and metric dependent. Each outer test fold contains roughly 300 benign rows, so a single false positive changes FPR by about 0.33 percentage point. The data cannot resolve production regions several orders of magnitude lower.

$U + H$ increases total model-plus-query time by about 7.93% relative to Uncertainty. XAI-Full increases it by 4.86%, and query computation accounts for approximately 5.42% of its reported model-plus-query time. These small absolute times apply to a 2,000-row experiment and omit analyst annotation and production feature extraction.

6.2.3 Answer to RQ2

Explanation-derived signals add useful information relative to random acquisition. Directional entropy

Table 7: Confirmatory active-learning performance, target attainment, and computation. Labels-to-target excludes validation labels and full-pool reference cost.

Policy	ROC-AULC	AP-AULC	Labels to target	Query, 10 rounds (s)	Model + query (s)
Random	0.928777	0.868350	584	0.0194	4.1947
Uncertainty	0.931272	0.881934	520	0.0191	4.1454
$U + H$	**0.933026**	**0.883336**	508	0.2459	4.4741
$U + V$	0.930377	0.880164	**488**	0.2298	4.3826
XAI-Full	0.932202	0.881638	492	0.2358	4.3471

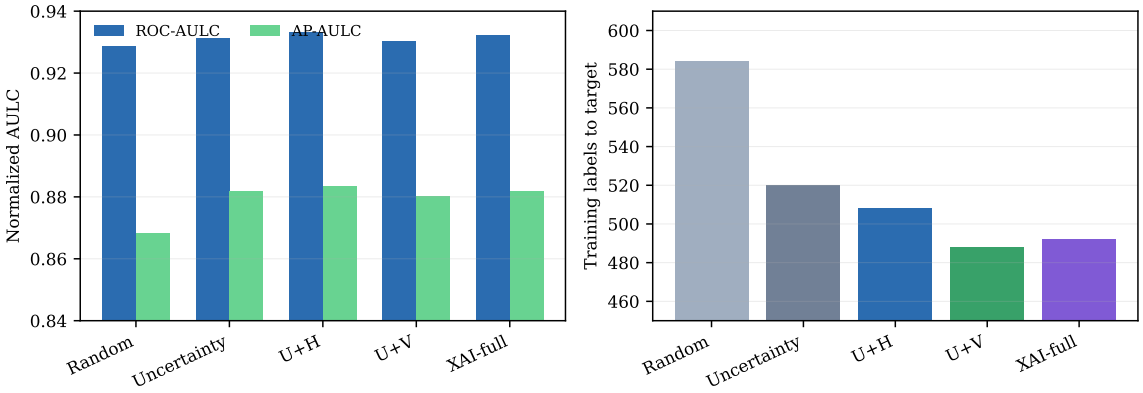


Figure 3: Confirmatory active-learning summaries. Higher AULC is better; fewer training labels to the predeclared validation target are better. Error bars are omitted because the source-of-truth report provides aggregate means and paired effects rather than fold vectors in this checkout.

Table 8: Selected paired outer-fold AL effects.

Comparison	ΔROC	ΔAP	Favorable folds
$U + H$ vs. Random	+0.004249	+0.014985	5/5
$U + H$ vs. Uncertainty	+0.001754	+0.001401	4/5
XAI-Full vs. Random	+0.003425	+0.013287	5/5
XAI-Full vs. Uncertainty	+0.000931	−0.000297	3/5 ROC

yields the strongest integrated learning trajectory; variance reaches the declared target with the fewest labels; and the full policy saves 92 training labels relative to Random. Improvements over uncertainty alone are smaller and change with the metric. The evidence supports objective-dependent sample prioritization, not a single dominant policy or a measured reduction in analyst hours.

6.3 RQ3: Quantitative TreeSHAP Validation

6.3.1 Controlled deletion

Table 10 reports AUC decreases over 25 fitted models. Top- |SHAP| deletion has a large monotonic effect as k increases, whereas uniform random deletion remains near zero. Under mean replacement at $k = 50$, the decreases are 0.306729 and 0.000979, an absolute difference of 0.305750. Median replacement gives a similar difference of 0.300119. The conclusion is robust to the two training-derived replacement baselines.

Equal coordinate count is an incomplete control. Uniform masks often select inactive tree features and do not match standardized displacement or manifold distance. The result nevertheless demonstrates

Table 9: Active-learning summaries in restricted false-positive regions.

Policy	pAUC, FPR$\leq$ 5%	TPR-AULC, FPR cap 1%	TPR-AULC, FPR cap 5%
Random	0.804096	0.580620	0.731980
Uncertainty	**0.826300**	**0.638180**	0.769260
$U + H$	0.824774	0.636780	0.770620
$U + V$	0.825194	0.626260	0.768140
XAI-Full	0.823525	0.631860	**0.771460**

Table 10: Controlled feature-space deletion. Values are ROC–AUC decreases from the corresponding unmasked model.

Replacement	k	Top-SHAP drop	Random drop	Top minus random
Mean	10	0.047817	0.000200	0.047617
Mean	25	0.135853	0.000530	0.135323
Mean	50	**0.306729**	0.000979	**0.305750**
Median	10	0.053483	0.000094	0.053388
Median	25	0.125180	0.000326	0.124854
Median	50	**0.300777**	0.000658	**0.300119**

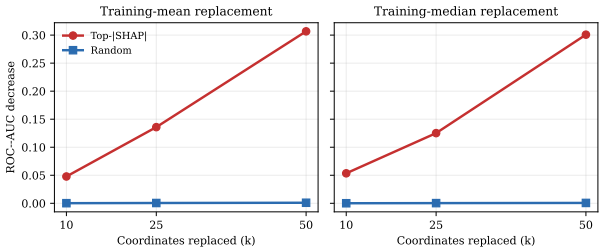


Figure 4: AUC decrease under top-|SHAP| and equal-count random replacement. Absolute effects are emphasized because a ratio is unstable when the random-control decrease is near zero.

relative fitted-model reliance: under both specified interventions, TreeSHAP ranks coordinates whose replacement changes ranking much more than uniformly chosen coordinates. It does not show a valid edit to a PE file or a causal malware mechanism.

6.3.2 Rank stability

Within a fixed fold, global Spearman agreement is 0.6144, and top-10 Jaccard agreement is 0.6221. Across folds, Spearman falls to 0.5512 and top-10 Jaccard to 0.4416. Agreement decreases further at larger top sets across folds. The ranking is therefore neither arbitrary nor invariant; the training partition changes it more than controlled model randomness.

Table 11: TreeSHAP ranking stability.

Comparison	ρ	J_{10}	J_{25}	J_{50}
Seed, same fold	0.6144	0.6221	0.5398	0.4901
Different fold	0.5512	0.4416	0.3073	0.3020

Mean EC is 0.4268 ± 0.0363 and mean CF is 0.4680 ± 0.0465 across 25 models. Maximum additivity error is approximately 6×10^{-15}. These values describe geometry and numerical consistency; they are not combined into a fidelity certificate.

6.3.3 Answer to RQ3

TreeSHAP top-ranked coordinates exert substantially greater model-ranking impact than equal-budget random coordinates under both mean and median replacement. Moderate stability supports repeatable model reliance while exposing meaningful partition sensitivity. The evidence is quantitative but model relative: it does not establish feature causal-

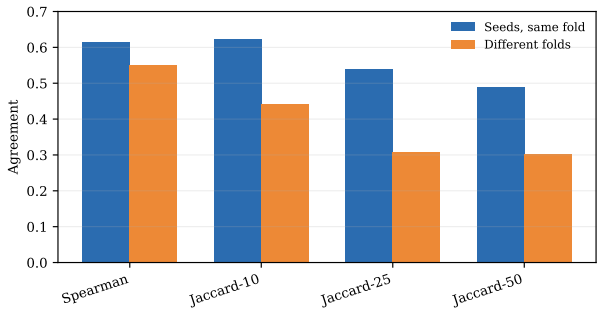


Figure 5: Global and top-k attribution agreement. Fold changes have a larger effect than seed changes within the same fold.

ity, executable-valid masking, or timeless malware semantics.

6.4 Exploratory Fusion Extensions

The 1,186-sample horizon-40 subset was used after observing the horizon-20 limitations and is therefore exploratory. Robust PID v2 reached AUC 0.9895 and AP 0.9911, equal at reported precision to Logistic stacking, with FPR 3.35% versus 3.68% and TPR 94.32% versus 94.84%. The AUC difference interval $[-0.0008, +0.0008]$ and FPR interval both include zero. Against the older PID full v2, Robust PID v2 reduces FPR by 1.33 percentage points with CI $[-2.66, -0.15]$. This supports an algorithm candidate, not independent confirmation.

A gated residual PID stress test under a controlled five-event delay reports FPR 2.61% versus 3.93% for Temporal stacking and TPR 94.49% versus 94.66%. It removes eight false positives, creates none, and loses one true positive; Holm-adjusted $p = 0.0312$. Three of five folds select zero gain, and event-AULC does not improve. This evidence is limited to the specified stress construction.

Finally, the timestamp pilot downloads reports for 20 selected samples and finds eight usable event sequences with 4,697 API–timestamp pairs. It demonstrates that wall-clock sequences can be parsed. With only two usable benign and six malware sequences, it cannot evaluate performance, latency, or the derivative term. These exploratory results define the next confirmatory experiment and do not change the main RQ1 conclusion.

7 Discussion

7.1 A Unified Flow without an Integrated Claim

The three components address one recurring security-learning problem: useful evidence is incomplete, arrives at different times, and has different costs. Static and dynamic scores allocate evidence to an online decision. Active learning allocates a label budget to model maintenance. TreeSHAP diagnostics allocate confidence to an explanation ranking used in that maintenance process. This flow is scientifically useful because it identifies the interfaces a future system must validate: paired channel identity, acquisition lineage, and explanation checks.

The flow is not itself an empirical interaction. MalBehavD-V1 does not contain the EMBER active-learning artifact, and EMBER does not contain the ordered dynamic observation required for fusion. Consequently, the study cannot answer whether explanation-guided retraining improves the fused detector, whether PID changes which samples should be queried, or whether one common threshold reduces end-to-end alert cost. A factorial experiment on one paired cohort remains necessary. Preserving this boundary makes the component results easier to audit and prevents a conceptual architecture from becoming a fabricated system score.

7.2 What the PID Translation Establishes

The control analogy provides a disciplined set of nested hypotheses. P asks whether current detector disagreement should alter the mixture. I asks whether persistent signed disagreement supplies additional information. D asks whether a change in disagreement across event prefixes predicts useful adjustment. The empirical result supports the first hypothesis and rejects the need for the latter two under

the measured horizon.

This outcome is plausible. The static score is constant within a sequence, while the dynamic score updates at short API prefixes. If successive prefixes change only a few events, the derivative term can be sparse or noisy. If one channel is systematically higher for a source, the integral may accumulate collection bias rather than sample-specific reliability. A proportional mapping can capture the dominant behavior with fewer states and lower implementation risk. Adaptive P's FPR and Brier improvements over Fixed make it a stronger current candidate than full PID for horizon 1–20.

The result also clarifies what PID cannot do. It cannot reveal behavior that the sandbox never observes. It cannot turn event count into elapsed time. It cannot guarantee calibration or stability from control theory alone. Sleep-evasion resistance would require real timestamped traces, controlled delayed behavior, and a specified observation policy. End-to-end latency would require separate measurement of queueing, sandbox startup, observation, feature extraction, channel inference, and fusion. The small timestamp pilot begins data-source validation but does not satisfy those requirements.

Learned stacking remains an important baseline. Full PID has comparable AUC and lower FPR at its own operating point, but also a small TPR reduction. Average has the best Brier and slightly higher AUC than the adaptive family. These results show that interpretability and operating-point objectives, rather than a tiny AUC difference, must motivate an adaptive controller. External validation at a common target TPR should decide whether Adaptive P's lower FPR transfers.

7.3 What the Acquisition Result Means

Directional entropy adds a small and repeatable learning-curve improvement over Random and a smaller improvement over Uncertainty in four of five folds. It distinguishes two probability-equivalent cases: weak evidence and stronger opposing signed masses. Such conflict can identify a model region where a new label is useful. The algebra in Equation (19) narrows this interpretation: H is determined by the total raw-score contribution and the L_1 attribution norm. A direct margin-plus-norm baseline may recover part of the effect and should be included in an extension.

No acquisition policy dominates every objective. $U+H$ leads AULC, $U+V$ reaches the target earliest, XAI-Full balances signals and saves 92 labels relative to Random, and Uncertainty remains strongest in two restricted-FPR summaries. Policy selection should therefore follow the maintenance objective. An organization prioritizing average model quality during iterative labeling may prefer $U+H$; one using the retained validation target may prefer $U+V$; one requiring low computation and strong low-FPR behavior may retain Uncertainty.

The label result has an explicit accounting boundary. It counts labels added to training. Validation labels, initial dataset creation, the full-pool reference, analyst abstention, and disagreement resolution remain outside the number. A 15.8% label-count reduction is not a measured 15.8% reduction in labor.

Batch redundancy is an unresolved alternative explanation. Selecting the top 100 pointwise scores can query many objects from one region. Uncertainty plus k-center selection would test whether simple diversity reproduces the gain. A calibrated committee would test whether ensemble disagreement explains the benefit. Robust quantile or rank normalization would test sensitivity to extreme H or V values that compress min–max scaling. These settings must be frozen within a nested design, not chosen by outer-test AULC.

7.4 What the TreeSHAP Audit Establishes

Top-ranked replacement damages model ranking much more than uniform random replacement under both training means and medians. This is strong evidence of relative model reliance under the defined feature-space intervention. It is particularly useful as an internal release check: a TreeSHAP ranking that fails to exceed equal-count random deletion should not be presented as a meaningful ordering of influential coordinates.

The deletion gap is not a causal certificate. Tree ensembles use a subset of 2,381 coordinates, so random masks often strike inactive features. Top-ranked coordinates may also be farther from the replacement baseline, creating a larger standardized displacement and greater off-manifold shift. Stronger controls should sample only used features, match displacement magnitude, replace semantically linked PE groups, and use conditional backgrounds. Reconstructing valid PE binaries would answer a different and substantially harder question.

Moderate stability gives a complementary warning. Explanations change less across model seeds in one fold than across folds, showing that data composition matters. A global top-feature list should therefore be versioned with the model, feature schema, and training cohort. For active learning, round-specific change is expected because the model is retrained, but abrupt attribution drift should be logged and investigated.

Additivity, EC, and CF remain diagnostics of numerical or geometric properties. A common permutation of attribution columns can preserve sums, cosines, norms, and centroid separation while giving effects the wrong feature identities. Combining such heterogeneous quantities into an unvalidated Overall Fidelity Score would hide this weakness. Separate reporting is more informative and more defensible.

7.5 Operational Interpretation

The most immediate use is a controlled model-development workflow. A candidate static model scores a pool; a chosen acquisition rule prioritizes objects for an authorized analyst; query/model lineage is retained; and deletion/stability checks are run before release. Separately, when a paired dynamic sequence is available, Adaptive P can be compared against Average, Fixed, and Stacking at a locked operating objective. The evidence does not yet authorize automated remediation or deployment claims.

Scaling introduces additional costs. TreeSHAP over a large candidate pool can be reduced by first filtering a probability band, but that changes the acquisition population and requires reevaluation. Dynamic collection dominates arithmetic fusion cost, so optimizing the PID calculation alone will not solve system latency. These practical distinctions follow directly from the unified evidence- allocation view: computation, labels, observation time, and alert errors are different resources and should be measured separately.

7.6 Evidence Needed for a Stronger Journal Claim

The highest-priority experiment is an external or temporal paired cohort whose method and thresholds are frozen before access. Adaptive P and robust PID v2 should be preregistered with a common-TPR comparison and noninferiority or superiority margins. Timestamped event sequences should support separate early- detection and wall-clock analyses. The same cohort should then receive a factorial acquisition–fusion evaluation if interaction is claimed.

For RQ2/RQ3, the strongest additions are uncertainty plus diversity, a calibrated committee, weight/normalization sensitivity, family- or time-aware splits, used-feature and magnitude-matched deletion controls, conditional/group masking, and a second explainer. These experiments address plausible alternatives to the current effects and determine how far they generalize beyond the audited subset.

8 Threats to Validity

8.1 Construct Validity

Both datasets encode generic benign/malware labels. Neither establishes ransomware membership, ransomware behavior, or zero-day status. AVClass metadata is incomplete and not used as behavioral ground truth. The paper therefore uses the term Windows malware and treats ransomware-specific detection as future work.

MalBehavD-V1 prefixes are ordered by API-event count. Event 20 is not 20 seconds, and a lower arithmetic fusion cost does not imply earlier wall-clock detection. Detector disagreement is a surrogate fusion signal, not the tracking error of a physical plant. Terms such as proportional, integral, and derivative

describe the mathematical update only; they do not confer closed-loop stability.

Directional entropy is entropy of positive/negative attribution mass, not predictive class entropy. Attribution variance is dispersion across coordinates of one SHAP vector, not epistemic variance. "Explanation-guided" describes the source of the acquisition signals, not proof that they encode human-understandable malware mechanisms.

Deletion measures fitted-model behavior under feature replacement. It does not establish causal features, valid PE transformations, adversarial robustness, or analyst usefulness. EC, CF, and additivity measure different constructs and are not aggregated into an Overall Fidelity Score.

8.2 Internal Validity

Fusion methods use paired static/dynamic rows and validation-selected settings, which reduces channel and threshold confounding. Yet the adaptive fusion design was developed after inspecting the original confirmatory result on the same 1,922-sample cohort. The resulting test-row estimates are development evidence; same-cohort bootstrap intervals cannot remove method-selection bias. Source differences between benign and malware data may also influence both channel scores despite exclusion of explicit verdict and threat-name metadata.

For AL, outer test partitions do not enter query selection or target definition, and policies share initial sets within a fold/seed. The fixed acquisition weights and min–max normalization were not selected through an independent outer sensitivity analysis. Pointwise top-100 selection has no diversity constraint. The simulated oracle returns retained labels without error, whereas analysts can abstain, disagree, or use different evidence.

For fidelity, mean and median baselines use training data, avoiding direct test leakage. Top and random masks are matched by coordinate count but not by standardized displacement, tree usage, semantic group, or manifold distance. These differences can inflate the top-minus-random gap. Masked test vectors are not verified against the PE format.

8.3 Statistical Conclusion Validity

Fusion confidence intervals are paired over retained predictions, but only one development cohort is available. Multiple metrics and method comparisons create selection opportunities, and no external confirmatory family controls them. FPR and TPR use validation-selected operating points; the PID–stacking FPR difference is not a comparison at equal TPR.

The AL comparison unit is the outer fold after averaging five seed trajectories. Five folds provide effect sizes and direction counts but limited asymptotic inference. Twenty-five fold–seed runs do not represent 25 independent datasets. The source report supplies aggregate values and selected paired effects rather than fold vectors in this checkout, so reconstructed figures omit uncertainty bars.

Restricted-FPR estimates use small test folds. With approximately 300 benign rows, one false positive changes FPR by roughly 0.33 percentage point. Results at 1% or 5% caps cannot validate alert rates at 10^{-3} or 10^{-4}.

The 25 fidelity models share folds and data; they are not independent population draws. Random masks characterize conditional mask variability. Pairwise stability values share fitted models and cannot be entered as independent rows in a naive significance test.

8.4 External Validity

The 2,000-row EMBER subset covers a short 2018 period and uses random row folds. It does not impose temporal, family, campaign, or source separation. The MalBehavD-V1 cohort is also one source with metadata-availability and minimum- event selection. Results can change for newer software, different organizations, other sandbox environments, or another feature extractor.

The horizon-40 subset was selected after horizon-20 results were known. Robust PID v2 and controlled stress findings are therefore explicitly exploratory. The timestamp pilot contains only eight usable sequences and cannot support performance inference. No measured result covers production queueing, an-

alyst workflow, or end-to-end incident response.

LightGBM and TreeSHAP are central to RQ2/RQ3. The acquisition summaries can be defined for other additive attributions, but their numeric behavior, cost, and stability may differ for raw-byte networks, graph models, or sequence models.

8.5 Integration Validity

The three RQs use two datasets and different feature spaces. The architecture connects them as stages of an auditable learning process, but the experiments do not demonstrate a single deployed pipeline. There is no measured gate deciding which EMBER sample receives dynamic analysis, no AL retraining of the MalBehavD-V1 fusion channels, and no shared latency/cost objective. Claims about synergy, interaction, or complete-system superiority are outside the evidence.

8.6 Reproducibility Validity

The source-of-truth report names commit `8bfa06b65b93`, dataset hashes, result directories, and verification artifacts. The remote branch available in the writing environment ends at an older commit, so the manuscript was generated from the supplied report rather than by rerunning the missing artifacts. Tables and plots reproduce reported aggregate values; they do not constitute an independent computational replication.

Before submission, the named commit and all result directories must be pushed, hashes checked, verification commands rerun in a clean environment, and tables and figures regenerated directly from fold-level records. Bibliographic metadata, data-sharing rights, authorship order, corresponding author, and the target journal template also require final verification. Computational replay can show consistency with stored code and data; it cannot prove upstream label accuracy or external generalization.

9 Conclusion

This paper formulated Windows malware model development as auditable evidence allocation under partial observability. The decision stage combines early static evidence with event-indexed dynamic evidence; the adaptation stage assigns a limited labeling budget; and the audit stage tests the attribution ranking used in model inspection and acquisition. The formulation provides one scientific flow while preserving component-specific estimands and datasets.

On the 1,922-sample MalBehavD-V1 development cohort, Adaptive P reduced FPR from 12.45% to 11.16% relative to fixed weighting and reduced Brier by 0.00306, with paired intervals excluding zero. Full PID obtained AUC comparable to logistic stacking and a lower FPR at its independently selected operating point, with a small TPR trade-off. Integral and derivative terms did not add consistent value over P within 1–20 event prefixes. The evidence therefore favors a simpler proportional adaptation and treats richer PID variants as hypotheses for external timestamped evaluation.

On the identity-audited 2,000-sample EMBER subset, $U + H$ produced the strongest ROC/AP learning-curve areas. $U + V$ reached the predeclared target with the fewest labels, while XAI-Full used 492 labels, 92 fewer than Random. Gains over Uncertainty were smaller and metric dependent. Across 25 fitted models, top-|SHAP| deletion produced much larger AUC decreases than equal- budget random deletion under mean and median replacement, while moderate stability revealed dependence on data partitions.

These results support component-wise adaptive fusion, explanation-guided sample prioritization, and quantitative attribution auditing for generic Windows malware. They do not establish ransomware detection, causal PE explanations, external generalization, or an end-to-end integrated system. The next evidence gate is a frozen external or temporal paired cohort with common operating targets and real timestamps, followed by stronger AL diversity baselines, conditional deletion controls, and a factorial acquisition–fusion experiment on one cohort.

A Fusion Replay and External-Validation Checklist

A submission-quality replay of RQ1 should retain, for every outer fold:

1. the cohort manifest with SHA-256, label, split, session/sequence ID, static metadata row, and ordered dynamic event identifiers;
2. proof that verdict, detection-count, threat-name, filename, and other direct label proxies are excluded from the static allow-list;
3. channel models, calibration objects, prefix scores, validation/test predictions, and the provenance of each score;
4. fixed-weight grid, stacking configuration, P/PI/PID settings, controller-state reset, clipping rule, and deterministic tie breaking;
5. validation-only selection records for gains, mixture settings, and thresholds, with no state transfer into held-out rows;
6. per-prefix and final predictions for Average, Max-risk, Fixed, Stacking, P, PI, and PID using identical paired inputs;
7. ROC–AUC, AP, Brier, FPR, TPR, confusion counts, and paired bootstrap indices sufficient to reproduce every interval;
8. separate wall-clock measurements for collection, feature extraction, channel inference, fusion arithmetic, queueing, and total response if a latency claim is introduced.

The external experiment should freeze Adaptive P and any robust PID candidate before data access. A common target TPR or a preregistered cost function should govern threshold comparison. Noninferiority and superiority margins must be declared before evaluation. Timestamped sequences should establish whether a derivative term sees meaningful elapsed-time change rather than event-count noise.

B Active-Learning Replay Checklist

For each outer fold and AL seed, retain:

1. development, validation, and test row identifiers;
2. the shared 200-row initial labeled set and ordered remaining pool;
3. model configuration, library versions, and deterministic seeds;
4. validation/test probabilities and labels at every round;
5. pool probabilities, $U/H/V$ components, normalized values, composite scores, ranks, and queried row identifiers;
6. the full-pool validation reference and target calculation;
7. per-round ROC–AUC, AP, low-FPR summaries, and scoped timing records;
8. checkpoints or a deterministic reconstruction mechanism;
9. verification of query membership, uniqueness, additivity, metric recomputation, and absence of validation/test leakage.

Every policy in a fold/seed must begin from the same initial set. The test partition must remain unavailable to query selection, target construction, threshold choice, weight selection, and stopping. A stronger extension should add uncertainty plus k-center diversity, a calibrated committee, direct margin-plus-attribution-norm controls, and frozen sensitivity grids for normalization and α, β, γ.

C TreeSHAP Audit Checklist

For each fold and model seed:

1. retain the fitted model, split IDs, feature schema, training mean and median, and raw TreeSHAP matrix with bias stored separately;

2. verify raw-score additivity before ranking features;
3. save top-k and random mask indices for every sample, k, control draw, and replacement baseline;
4. reconstruct masked matrices from original test vectors and saved indices, rather than from plotted summaries;
5. retain original and masked probabilities and recompute AUC decreases;
6. retain global rankings and model-pair identifiers used for Spearman and Jaccard summaries;
7. label EC/CF as geometry diagnostics and confirm that no composite OFS is introduced without validated scaling, weights, and sensitivity analysis.

Required stronger controls include random masks restricted to model-used features, standardized-displacement matching, attribution-column permutation, schema-aware group replacement, conditional backgrounds, and comparison with at least one alternative explainer or importance method.

D Metric and Interpretation Boundaries

E Claim–Evidence Map

F Artifact Traceability

The submission package should map each paper object to the following retained families:

- **EMBER identity:** `results/confirmatory_20261019_ember_identity/static_identity_audit.json` and `SHA256SUMS`.
- **Active learning:** `al/aulc_summary.csv`, `target_reach_summary.csv`, `fold_level_effects.csv`, `low_fpr_aulc_by_fold.csv`, `overhead_by_fold.csv`, and `verification.json`.
- **Fidelity:** `fidelity/deletion_summary.csv`, `shap_stability_summary.csv`, `model_deletion_comparisons.csv`, and `verification.json`.
- **Fusion confirmatory:** `results/measured_20261023_malbehavd_pid_confirmatory_1922/` including cohort, source, summary, D-ablation, paired-bootstrap, and verification records.
- **Fusion development:** `results/measured_20261028_fusion_only_evidence_1922/` including `fusion_only_summary.json`.
- **Exploratory fusion:** `results/exploratory_20261028_pid_v2_meta_fusion_horizon40/` and the stateful/stress result families.
- **Timestamp feasibility:** `results/measured_20261028_vt_behaviour_pilot20/`.

The machine-readable companion file `CLAIM_EVIDENCE_MAP.csv` expands this mapping with permitted manuscript locations and exact qualifiers.

G Pre-Submission Evidence Gates

Before journal submission:

1. synchronize source commit `8bfa06b65b93` and every artifact named above; run hash and replay verification in a clean environment;
2. regenerate tables, confidence intervals, and figures directly from fold-level records, including uncertainty visualization;
3. freeze Adaptive P/robust PID and run a new external or temporal cohort at a preregistered common operating target;
4. add uncertainty-plus-diversity, a calibrated committee, and acquisition-weight/normalization sensitivity;

Table 12: What the reported quantities establish and do not establish.

Quantity	Supported interpretation	Unsupported interpretation
Fusion AUC/AP	Ranking of paired cohort scores	Deployment utility or external transfer
FPR/TPR	Decisions at validation-selected thresholds	Same-TPR comparison when thresholds differ
Brier	Squared probabilistic error on the cohort	Complete calibration characterization
P improvement	Value of current disagreement in measured development cohort	Classical-control stability or universal PID value
I/D ablation	No stable incremental benefit in tested prefixes	Proof that temporal memory can never help
ROC/AP-AULC	Ranking quality integrated over measured label budgets	Analyst labor, calibration, or external generalization
Labels to target	Training labels to a fold-specific validation target	Total annotation cost or hours saved
Low-FPR AL summaries	Relative behavior at 1%/5% caps in outer folds	Reliable FPR at 10^{-3} or 10^{-4}
H	Balance of positive/negative TreeSHAP mass	Predictive entropy or causal conflict
V	Dispersion across signed attribution coordinates	Ensemble or posterior uncertainty
Top-k deletion	Fitted-model sensitivity under training-statistic replacement	Valid PE transformation or causal malware mechanism
Additivity	Numerical reconstruction of the raw tree score	Correct feature identity or human usefulness
Seed/fold agreement	Sensitivity to fitted model and data partition	Stability across organizations and time
Replay verification	Consistency with retained code/data/artifacts	Independent replication or upstream data authenticity

5. add identity-sensitive, used-feature, magnitude-matched, and group-aware fidelity controls;

6. obtain temporal, family-disjoint, or source-disjoint evidence if the target venue expects broad generalization;

7. verify all bibliography metadata against publisher records and confirm authorship, ORCID, corresponding author, data rights, funding, conflicts, and the current journal template.

Table 13: Central claims, evidence tier, and manuscript status.

Claim	Status/tier	Required scope
Adaptive P lowers FPR and Brier versus Fixed	Supported, development	1,922-sample same-cohort development; paired intervals exclude zero
Full PID outperforms all fusion baselines	Not supported	Average leads AUC/Brier; P leads FPR; PID has TPR trade-off
I/D improve fusion	Not supported	No stable incremental benefit over P at 1–20 prefixes
Robust PID v2 matches stacking	Exploratory	Horizon-40 development subset; external confirmation pending
$U+H$ exceeds Random in ROC/AP-AULC	Supported, confirmatory	Five outer folds; paired effect and direction reported
$U+H$ universally exceeds Uncertainty	Not supported	Small effects; favorable in four of five folds
XAI-Full saves 92 training labels versus Random	Supported, confirmatory	Validation/reference and analyst cost are additional
XAI-Full is best on every criterion	Not supported	$U+H$ leads AULC; $U+V$ uses fewest labels
Top-SHAP deletion exceeds random deletion	Supported, confirmatory	Feature-space intervention; 25 models; mean/median
TreeSHAP identifies causal malware features	Not supported	No executable-valid or causal intervention
The complete architecture is validated end-to-end	Not supported	Two datasets and three component-level estimands
The study detects ransomware or zero-day malware	Not supported	Generic Windows malware labels only

References

[1] Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[2] Hyrum S. Anderson and Phil Roth. EMBER: An open dataset for training static PE malware machine learning models. *arXiv preprint arXiv:1804.04637*, 2018.

[3] Daniel Arp, Erwin Quiring, Feargus Pendlebury, Alexander Warnecke, Fabio Pierazzi, Christian Wressnegger, Lorenzo Cavallaro, and Konrad Rieck. Dos and don'ts of machine learning in computer security. In *31st USENIX Security Symposium*, pages 3971–3988, 2022.

[4] Jordan T. Ash, Chicheng Zhang, Akshay Krishnamurthy, John Langford, and Alekh Agarwal. Deep batch active learning by diverse, uncertain gradient lower bounds. In *International Conference on Learning Representations*, 2020.

[5] Karl J. Åström and Tore Hägglund. The future of PID control. *Control Engineering Practice*, 9(11):1163–1175, 2001.

[6] Alessandro Bacci, Alberto Bartoli, Fabio Martinelli, Eric Medvet, and Francesco Mercaldo. Detection of ransomware attacks through machine learning techniques. *Computers & Security*, 104:102213, 2021.

[7] William H. Beluch, Tim Genewein, Andreas Nürnberger, and Jan M. Köhler. The power of ensembles for active learning in image classification. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 9368–9377, 2018.

[8] David A. Cohn, Zoubin Ghahramani, and Michael I. Jordan. Active learning with statistical models. *Journal of Artificial Intelligence Research*, 4:129–145, 1996.

[9] Bhavya Ghai, Q. Vera Liao, Yunfeng Zhang, Rachel Bellamy, and Klaus Mueller. Explainable active learning (XAL): An empirical study of how local explanations impact annotator experience. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 2021.

[10] Daniel Gibert, Carles Mateu, and Jordi Planes. The rise of machine learning for detection and classification of malware: Research developments, trends and challenges. *Journal of Network and Computer Applications*, 153:102526, 2020.

[11] Richard Harang and Ethan M. Rudd. SOREL-20M: A large scale benchmark dataset for malicious PE detection. In *Proceedings of the 13th ACM Workshop on Artificial Intelligence and Security*, 2020.

[12] Sara Hooker, Dumitru Erhan, Pieter-Jan Kindermans, and Been Kim. A benchmark for interpretability methods in deep neural networks. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[13] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, pages 3146–3154, 2017.

[14] Amin Kharraz, William Robertson, Davide Balzarotti, Leyla Bilge, and Engin Kirda. Cutting the gordian knot: A look under the hood of ransomware attacks. In *Detection of Intrusions and Malware, and Vulnerability Assessment*, pages 3–24. Springer, 2015.

[15] Andreas Kirsch, Joost van Amersfoort, and Yarin Gal. BatchBALD: Efficient and diverse batch acquisition for deep bayesian active learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[16] Scott M. Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M. Prutkin, Bala

Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2:56–67, 2020.

[17] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, pages 4765–4774, 2017.

[18] Edward Raff, Jon Barker, Jared Sylvester, Robert Brandon, Bryan Catanzaro, and Charles Nicholas. Malware detection by eating a whole EXE. In *AAAI Workshop on Artificial Intelligence for Cyber Security*, 2018.

[19] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1135–1144, 2016.

[20] Joshua Saxe and Konstantin Berlin. Deep neural network based malware detection using two dimensional binary program features. In *10th International Conference on Malicious and Unwanted Software*, pages 11–20, 2015.

[21] Ozan Sener and Silvio Savarese. Active learning for convolutional neural networks: A core-set approach. In *International Conference on Learning Representations*, 2018.

[22] Burr Settles. Active learning literature survey. Technical Report 1648, University of Wisconsin–Madison, Department of Computer Sciences, 2009.

[23] Daniele Sgandurra, Luis Muñoz-González, Rabih Mohsen, and Emil C. Lupu. Automated dynamic analysis of ransomware: Benefits, limitations and use for detection. *arXiv preprint arXiv:1609.03020*, 2016.

[24] Michael Sikorski and Andrew Honig. *Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software*. No Starch Press, San Francisco, CA, USA, 2012.

[25] Dylan Slack, Sophie Hilgard, Emily Jia, Sameer Singh, and Himabindu Lakkaraju. Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, pages 180–186, 2020.

[26] Daniele Ucci, Leonardo Aniello, and Roberto Baldoni. Survey of machine learning techniques for malware analysis. *Computers & Security*, 81:123–147, 2019.

[27] Alexander Warnecke, Daniel Arp, Christian Wressnegger, and Konrad Rieck. Evaluating explanation methods for deep learning in security. In *2020 IEEE European Symposium on Security and Privacy*, pages 158–174, 2020.

[28] Chih-Kuan Yeh, Cheng-Yu Hsieh, Arun Sai Suggala, David I. Inouye, and Pradeep Ravikumar. On the (in)fidelity and sensitivity of explanations. In *Advances in Neural Information Processing Systems*, volume 32, 2019.